

SARVESH MAHALINGAM

Melbourne, VIC, Australia | +61 439 238 165 | sarveshmahalingam2004@gmail.com
[linkedin.com/in/sarvesh-mahalingam-459084221](https://www.linkedin.com/in/sarvesh-mahalingam-459084221) | github.com/SarveshMahalingam

PROFESSIONAL SUMMARY

Master of Data Science candidate at Monash University with hands-on experience architecting end-to-end data pipelines, building multi-agent AI systems, and delivering data-driven insights to technical and non-technical stakeholders alike. Skilled in Python, SQL, LangGraph, and the Databricks platform (Delta Lake, Unity Catalog, PySpark), with a track record of turning raw, real-world data into reliable, decision-ready analytics and agentic systems. Seeking an internship or part-time role in Data Science / AI Engineering, applying this experience to real infrastructure and sustainability challenges.

TECHNICAL SKILLS

Languages & Platforms: Python, SQL, Databricks (Delta Lake, Unity Catalog, PySpark)

AI & Machine Learning: LangGraph, scikit-learn, TensorFlow/Keras, LightGBM, spaCy

Tools & Cloud: AWS, Docker, Git/GitHub, Claude

EDUCATION

Monash University — Master of Data Science, WAM 84.25

Feb. 2026 – Present

VIT Chennai — B.Tech Computer Science Engineering, CGPA 8.57/10

Sep. 2021 – Sep. 2025

PROFESSIONAL EXPERIENCE

AI Engineer Intern — Infosys (Remote)

Jan. 2026 – Apr. 2026

- Led a 5-person cross-functional team through the full delivery lifecycle of an AI-powered application (**Streamlit, Gemini API**), coordinating workstreams and stakeholder communication to ship **15 days** ahead of schedule.
- Built a data ingestion and preprocessing pipeline (**BeautifulSoup, PyPDF**) with adaptive chunking logic, transforming **500+ page** unstructured documents into structured, high-fidelity inputs for downstream analysis.
- Designed a multi-format query engine and a **MongoDB**-backed analytics dashboard to surface structured insights from processed data, cutting deployment time by **25%** via **Docker** containerization.

PROJECTS & RESEARCH

HydroMesh — Multi-Agent Water Infrastructure Digital Twin — Python, LangGraph, wntr/EPANET, Streamlit

- Designed and built a five-agent **LangGraph** workflow simulating leak detection and mains-burst response for a water utility digital twin modeled on Yarra Valley Water (Melbourne), with agents for smart-meter monitoring, acoustic/pressure anomaly detection, hydraulic optimization, field dispatch, and customer communications.
- Ran deterministic hydraulic simulations (**wntr/EPANET**) for pipe-burst and PRV-throttling scenarios against a fire-service minimum-pressure safety check, keeping all agent decisions rule-based and reproducible offline, with an LLM used only to narrate agent reasoning for the dashboard.
- Built a **Streamlit** live-map dashboard and a **pytest** suite covering telemetry generation, hydraulic simulation, spatial dispatch logic, and full multi-agent workflow routing across three incident scenarios.

AEMO Energy Demand Forecasting Pipeline — Databricks, Delta Lake, Unity Catalog, PySpark, SQL

- Architected and built an end-to-end medallion (bronze/silver/gold) data pipeline on **Databricks (Unity Catalog, Delta Lake, serverless Spark SQL)**, ingesting and transforming ~4 years of half-hourly time-series data from two independent external sources into an analytics-ready feature store.
- Engineered **25+** calendar, weather, and time-series features (lag and rolling-window statistics, public-holiday flags, time-of-day classification) using advanced Spark SQL window functions, with independent automated verification (a second, structurally different recomputation in **pandas**) that every rolling/lag feature only references historical — never current or future — data.
- Authored a SQL-based business analytics suite translating **8** stakeholder-style business questions (cost-to-serve modeling, price-risk ranking, demand-weather relationships, revenue-opportunity sizing) into production SQL against the gold-layer warehouse, with accompanying **Databricks** dashboards/visualizations.

DDoS Attack Detection Model — Python, Keras/TensorFlow, scikit-learn

- Cleaned and preprocessed a **12.8M-row** network flow dataset (CICFlowMeter-style features), dropping near-constant/high-missingness columns and downcasting to **float32** to cut memory footprint for large-scale training.
- Performed correlation-based feature selection ($|r| > 0.95$) via heatmap analysis to reduce dimensionality, then trained a fully-connected network (Dense-512 → Dropout(0.4) → Dense-1, **Keras/TensorFlow**) for binary DDoS classification, achieving **99.04%** test accuracy on an 80/20 held-out split.

AI-Generated Text Detection System — Python, scikit-learn, spaCy, TF-IDF, Keras

- Engineered a **13-feature** stylometric/linguistic feature set (readability scores, spaCy POS-tag distributions, misspelling ratio, named-entity counts) alongside **34,000+** dimension TF-IDF vectors to detect AI-generated responses.
- Built and benchmarked an ensemble combining a calibrated **LinearSVC** (TF-IDF pathway) with a **Random Forest/LightGBM** classifier (linguistic-feature pathway) and a multi-input **Keras** network fusing both, reaching a final **ROC-AUC of 0.956** (vs. 0.932 sparse-only, 0.951 dense-only).

CERTIFICATIONS

- AWS Certified Solutions Architect
- Databricks Certified Data Analyst Associate